# Diffusion-Guided Spatiotemporal Cloud Motion Prediction

# Using Geostationary Infrared Satellite Sequences from GOES-16

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**Abstract**

Cloud motion prediction is an essential component of short-term weather forecasting and atmospheric monitoring. Traditional cloud forecasting techniques primarily rely on optical flow estimation or numerical weather prediction models that assume smooth motion propagation between frames. However, cloud systems often undergo complex structural changes such as deformation, formation, and dissipation, which makes accurate prediction challenging.

This paper proposes AURORA, a hybrid deep learning architecture designed for cloud motion prediction using geostationary satellite infrared imagery. The proposed system integrates transformer-based spatiotemporal feature extraction, optical flow motion estimation, Convolution-LSTM temporal modelling, and diffusion-based generative refinement. The model captures both cloud displacement and structural evolution to generate realistic future cloud states. Performance is evaluated using image similarity metrics such as Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Mean Squared Error (MSE). Experimental analysis demonstrates that the hybrid architecture improves structural fidelity and temporal consistency of predicted cloud patterns compared to traditional motion-based forecasting approaches.

## Introduction

Cloud monitoring and prediction play a critical role in meteorology, aviation safety, climate research, and disaster management. Modern weather forecasting systems rely heavily on satellite observations to monitor atmospheric conditions in near real-time.

Geostationary satellites such as GOES-16 continuously capture infrared imagery of Earth's atmosphere using the Advanced Baseline Imager (ABI). These infrared images provide information about cloud-top temperature and atmospheric dynamics, enabling meteorologists to observe cloud formation and movement.

Despite the availability of high-resolution satellite imagery, predicting future cloud states remains a difficult task. Atmospheric processes are highly nonlinear and involve complex interactions between temperature, pressure, and humidity. Clouds frequently change shape, merge with other cloud systems, or dissipate entirely.

Traditional cloud motion forecasting techniques rely on optical flow estimation or correlation tracking methods. While these approaches can estimate motion vectors between consecutive frames, they often fail to capture structural changes in cloud patterns.

Recent advances in deep learning and generative modeling provide new opportunities to improve cloud prediction systems. Diffusion models, transformers, and temporal neural networks have demonstrated strong capabilities in image generation and video prediction tasks.

Motivated by these advances, this work proposes AURORA, a diffusion-guided spatiotemporal architecture designed to predict future cloud states from sequences of satellite imagery.

## Problem Statement

Given a sequence of satellite infrared images captured at regular time intervals, the objective is to predict the future cloud state.

Let the sequence of past satellite frames be

$$X = \{I_{t-k}, I_{t-k+1}, \ldots, I_t\}$$

where

- $I_t$represents the satellite image at time $t$
- $k$represents the number of past frames used for prediction.

The objective is to estimate the next future frame

$$\hat{I}_{t+1}$$

The prediction function can be defined as

$$\hat{I}_{t+1} = F(I_{t-k}, \ldots, I_t)$$

where $F(\cdot)$represents the AURORA prediction model.

The challenge is to design a model capable of capturing both

- cloud displacement
- cloud structural evolution

from the given satellite image sequence.

**Research Overview**

This work investigates whether integrating motion estimation with generative deep learning models can improve the realism and accuracy of cloud motion prediction.

The proposed architecture combines the following components:

- Spatiotemporal feature extraction using transformers
- Optical flow motion estimation
- Temporal sequence modelling using Convolution-LSTM
- Diffusion-based generative refinement
- Pixel-wise fusion using attention networks

By integrating these modules, the system attempts to model both the motion dynamics and structural evolution of clouds.

**Dataset Description**

The dataset used in this study consists of infrared satellite imagery captured by the GOES-16 geostationary satellite.

Specifically, Band-13 infrared imagery (10.3 μm wavelength) is used, which represents cloud-top temperature and provides clear visibility of cloud structures.

Key dataset characteristics:

- Temporal resolution: approximately 10 minutes
- Spatial resolution: high-resolution infrared imagery
- Data format: grayscale temperature maps

Preprocessing steps include:

- Radiometric normalization
- Image resizing
- Tiling into 256 × 256 patches
- Temporal sequence generation

These sequences serve as inputs to the prediction model.

**Methodology**

Data Pre-Processing

Satellite images are normalized to ensure consistent intensity ranges. Each frame is divided into smaller patches to reduce computational complexity during training.

Optical Flow Motion Estimation

Optical flow estimates the motion between two consecutive frames.

Brightness constancy assumption:

$$I(x, y, t) = I(x + u, y + v, t + 1)$$

where

- $u$is horizontal motion
- $v$is vertical motion

Using Taylor expansion, the optical flow constraint equation becomes

$$\frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v + \frac{\partial I}{\partial t} = 0$$

The displacement field is defined as

$$F(x, y) = (u(x, y), v(x, y))$$

This provides coarse motion guidance for cloud movement.

**Spatiotemporal Feature Extraction**

Satellite images are divided into patches and embedded into feature vectors. A transformer architecture captures spatial relationships using self-attention.

Attention mechanism:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where

- $Q$= query matrix
- $K$= key matrix
- $V$= value matrix

This allows the network to capture global spatial dependencies across cloud patterns.

**Temporal Modelling with Convolution-LSTM**

Convolution-LSTM captures temporal dependencies across satellite frames.

Input gate:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1})$$

Forget gate:

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1})$$

Cell update:

$$C_t = f_t \odot C_{t-1} + i_t \odot tanh(W_{xc} * X_t)$$

Hidden state:

$$H_t = o_t \odot tanh(C_t)$$

Convolution-LSTM enables the network to learn evolving cloud dynamics.

**Diffusion Model for Cloud Generation**

Diffusion models generate images by gradually removing noise.

Forward diffusion process:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t} x_{t-1}, \beta_t I)$$

After multiple diffusion steps:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon$$

The neural network learns to reverse this process.

Training loss:

$$L = E_{t,x_0,\epsilon}[\| \epsilon - \epsilon_\theta(x_t, t) \|^2]$$

This enables generation of realistic cloud textures.

**Prediction Fusion**

Predictions from multiple modules are combined.

$$I_{pred} = w_1 I_{flow} + w_2 I_{ConvLSTM} + w_3 I_{diffusion}$$

where

$$w_1 + w_2 + w_3 = 1$$

The weights are learned using attention mechanisms.

**Evaluation Metrics**

Model performance is evaluated using the following metrics.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (I_i - \hat{I}_i)^2$$

Structural Similarity Index

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Peak Signal-to-Noise Ratio

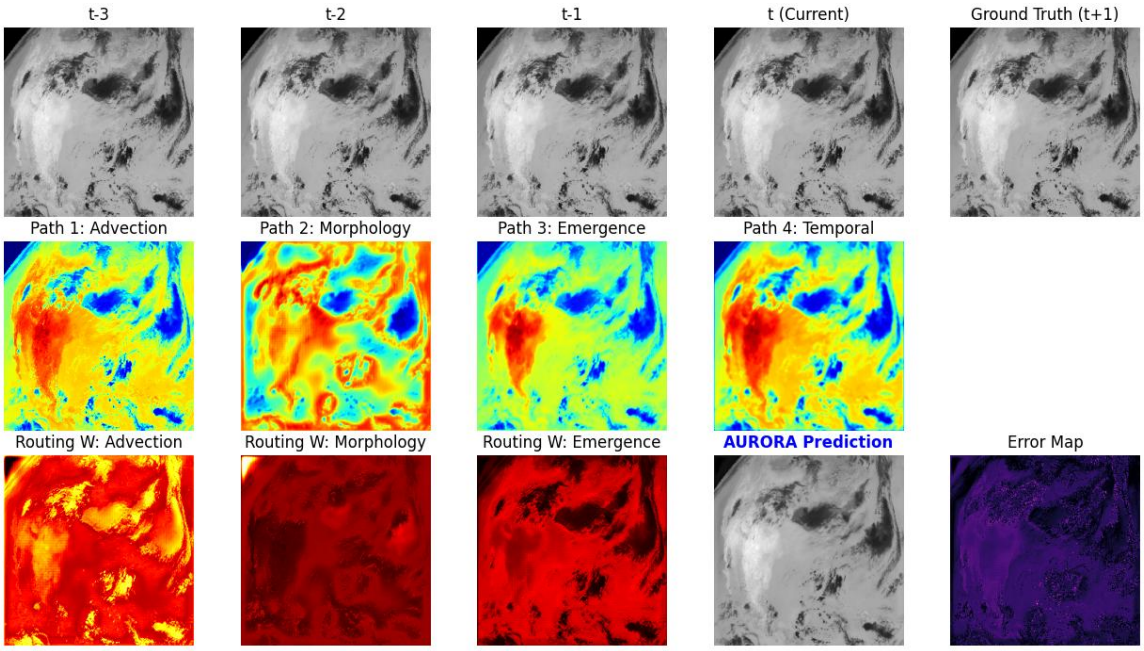
$$PSNR = 10\log_{10}\left(\frac{MAX^2}{MSE}\right)$$

These metrics evaluate prediction accuracy and structural fidelity.

## Experimental Results

The proposed architecture is evaluated using sequences of satellite infrared imagery.

The hybrid model demonstrates improved preservation of cloud boundaries and structural features compared to optical flow-only methods. Diffusion-based refinement significantly improves the visual realism of predicted cloud textures.




## Discussion

The results suggest that combining motion estimation with generative modelling improves cloud prediction accuracy. Traditional optical flow models capture displacement but fail to model structural cloud evolution.

The integration of Convolution-LSTM and diffusion models enables the system to learn temporal cloud dynamics and realistic cloud texture generation simultaneously.

However, training such hybrid architectures requires substantial computational resources.

## Limitations

Several limitations exist in the current implementation:

• Limited dataset size
• High computational cost of diffusion models
• Absence of atmospheric physics constraints
• Single-step prediction

## Future Work

Future improvements may include:

• Multi-step cloud forecasting
• Integration of multi-spectral satellite bands
• Real-time inference systems
• Physics-informed neural networks
• Large-scale global satellite datasets

## Conclusion

This work presented AURORA, a diffusion-guided spatiotemporal architecture for predicting cloud motion using geostationary satellite imagery. The proposed system integrates transformer-based feature extraction, optical flow motion estimation, temporal modelling, and diffusion-based generative refinement.

The hybrid design captures both cloud displacement and structural evolution, enabling the generation of more realistic and temporally consistent cloud predictions.

The results demonstrate that combining modern generative AI models with spatiotemporal learning techniques can significantly improve short-term weather forecasting systems.